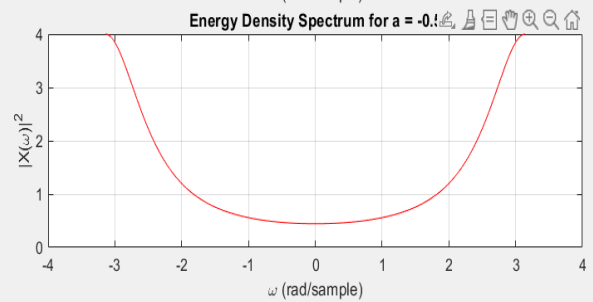
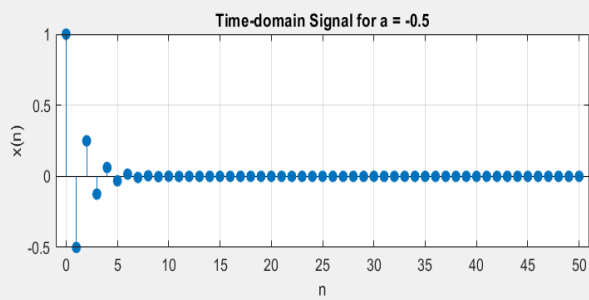
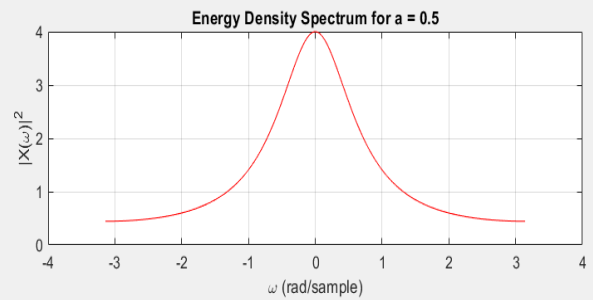
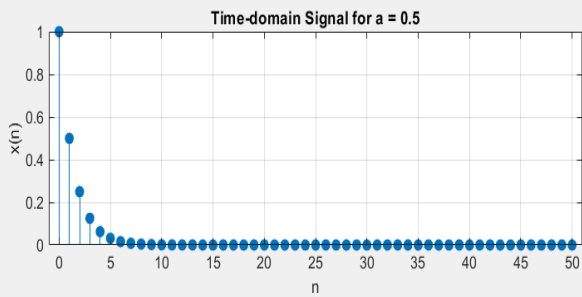
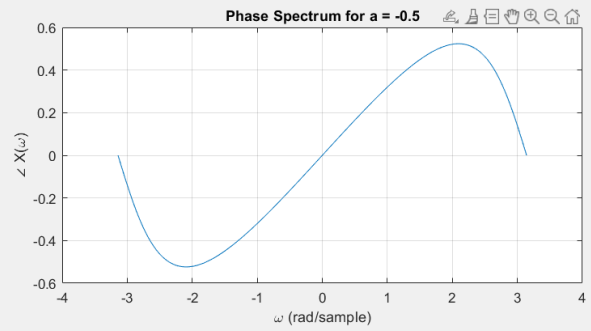
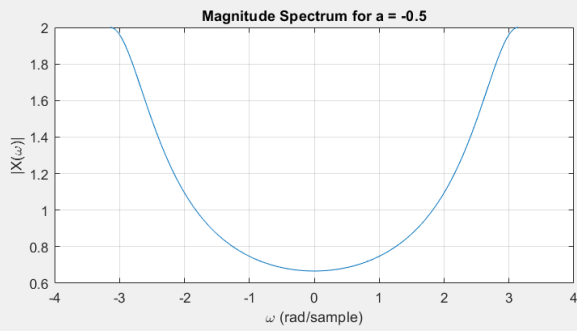
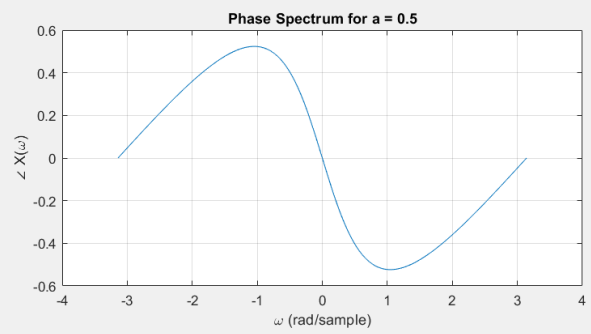
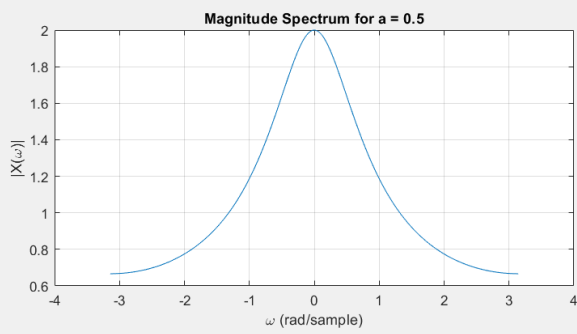


## EXERCISE#01

```
Editor - C:\Users\hasee\OneDrive\Documents\MATLAB\DSPLAB5\DTFT.m
DTFT.m EXERCISE2.m EXERCISE3.m EXERCISE4.m
1 function y = DTFT(x, n, w)
2     y = zeros(size(w));
3     for k = 1:length(w)
4         y(k) = sum(x .* exp(-1j * w(k) * n));
5     end
6 end
```

## EXERCISE#02

```
Editor - C:\Users\hasee\OneDrive\Documents\MATLAB\DSPLAB5\EXERCISE2.m
DTFT.m EXERCISE2.m EXERCISE3.m EXERCISE4.m
1 n = 0:50;
2 w = linspace(-pi, pi, 500);
3 a_values = [0.5, -0.5];
4
5 figure;
6 for i = 1:length(a_values)
7     a = a_values(i);
8     x = a.^n;
9     Xw = DTFT(x, n, w);
10    Sxx = abs(Xw).^2;
11
12    subplot(3,2,2*i-1);
13    stem(n, x, 'filled');
14    xlabel('n'); ylabel('x(n)');
15    title(['Time-domain Signal for a = ', num2str(a)]);
16    grid on;
17
18    subplot(3,2,2*i);
19    plot(w, Sxx, 'r');
20    xlabel('\omega (rad/sample)');
21    ylabel('|x(\omega)|^2');
22    title(['Energy Density Spectrum for a = ', num2str(a)]);
23    grid on;
24 end
25 figure;
26 for i = 1:length(a_values)
27     a = a_values(i);
28     x = a.^n;
29     Xw = DTFT(x, n, w);
30
31    subplot(2,2,2*i-1);
32    plot(w, abs(Xw));
33    xlabel('\omega (rad/sample)');
34    ylabel('|x(\omega)|');
35    title(['Magnitude Spectrum for a = ', num2str(a)]);
36    grid on;
37
38    subplot(2,2,2*i);
39    plot(w, angle(Xw));
40    xlabel('\omega (rad/sample)');
41    ylabel('angle X(\omega)');
42    title(['Phase Spectrum for a = ', num2str(a)]);
43    grid on;
44 end
```



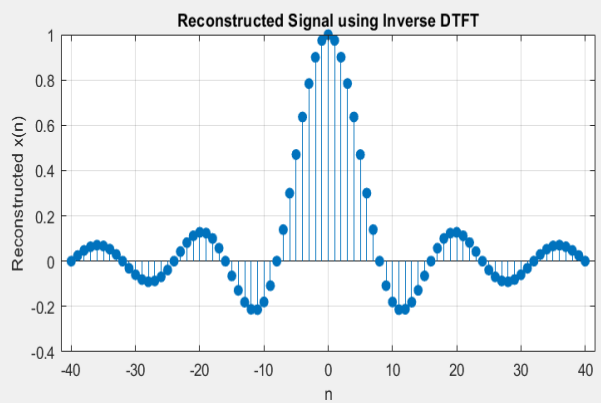
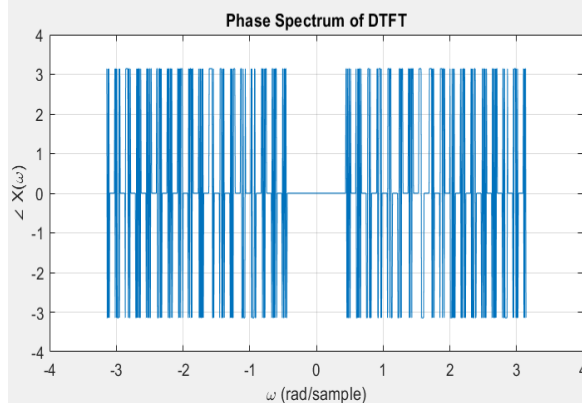
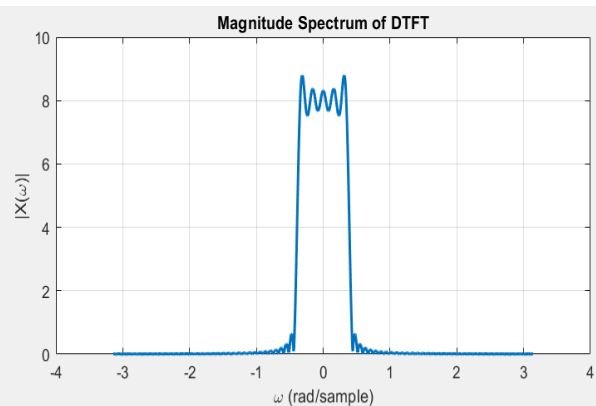
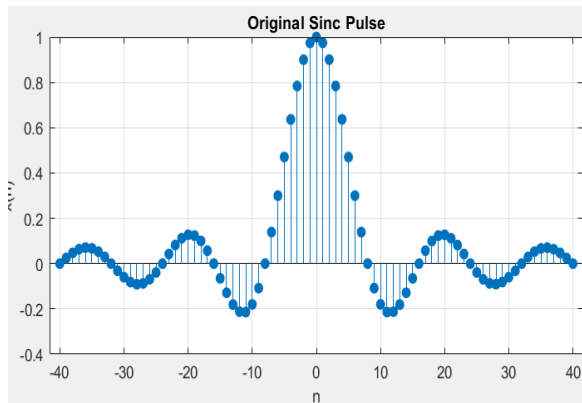
## EXERCISE#03

```

Editor - C:\Users\hasee\OneDrive\Documents\MATLAB\DSPLAB5\EXERCISE3.m
DTFT.m  EXERCISE2.m  EXERCISE3.m  EXERCISE4.m  EXERCISE6.m

1  n = -40:40;
2  f = 1/8;
3  x = sinc(f * n);
4
5  w = linspace(-pi, pi, 1000);
6  Xw = zeros(size(w));
7
8  for k = 1:length(w)
9      Xw(k) = sum(x .* exp(-1j * w(k) * n));
10 end
11
12 [W, N] = meshgrid(w, n);
13 dw = w(2) - w(1);
14 x_reconstructed = real(trapz(w, (1/(2*pi)) * (Xw .* exp(1j * W .* N)), 2));
15 figure;
16
17 subplot(2,2,1);
18 stem(n, x, 'filled');
19 xlabel('n'); ylabel('x(n)');
20 title('Original Sinc Pulse');
21 grid on;
22
23
24 subplot(2,2,2);
25 plot(w, abs(Xw), 'LineWidth', 1.5);
26 xlabel('\omega (rad/sample)'); ylabel('|X(\omega)|');
27 title('Magnitude Spectrum of DTFT');
28 grid on;
29
30 subplot(2,2,3);
31 plot(w, angle(Xw), 'LineWidth', 0.5);
32 xlabel('\omega (rad/sample)'); ylabel('\angle X(\omega)');
33 title('Phase Spectrum of DTFT');
34 grid on;
35
36
37 subplot(2,2,4);
38 stem(n, x_reconstructed, 'filled');
39 xlabel('n'); ylabel('Reconstructed x(n)');
40 title('Reconstructed Signal using Inverse DTFT');
41 grid on;
42

```

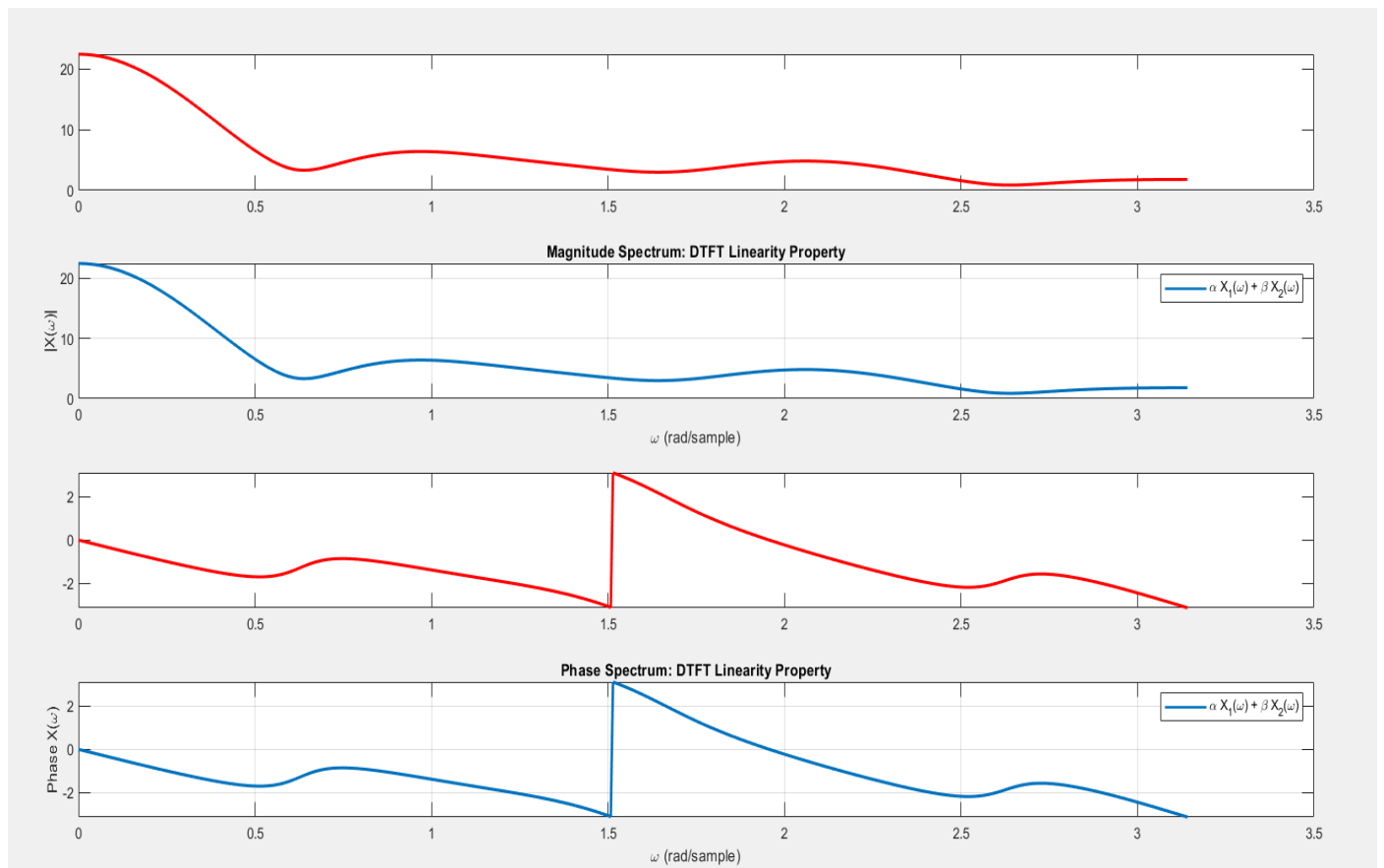


## EXERCISE#04

```

DTFT.m x EXERCISE2.m x EXERCISE3.m x EXERCISE4.m x EXERCISE6.m x EXERCISE7.m x POSTLAB.m x +
27 plot(w, abs(X_sum_time), 'LineWidth', 1.5);
28 xlabel('\omega (rad/sample)'); ylabel('|X(\omega)|');
29 title('Magnitude Spectrum: DTFT Linearity Property');
30 legend('\alpha X_1(\omega) + \beta X_2(\omega)', 'DTFT of (\alpha x_1(n) + \beta x_2(n))');
31 grid on;
32 subplot(4,1,3);
33 plot(w, angle(X_sum_freq), 'r', 'LineWidth', 1.5); hold on;
34 subplot(4,1,4);
35 plot(w, angle(X_sum_time), 'LineWidth', 1.5);
36 xlabel('\omega (rad/sample)'); ylabel('Phase X(\omega)');
37 title('Phase Spectrum: DTFT Linearity Property');
38 legend('\alpha X_1(\omega) + \beta X_2(\omega)', 'DTFT of (\alpha x_1(n) + \beta x_2(n))');
39 grid on;
40
41
42 function Xw = dtft(x, n, w)
43     Xw = zeros(size(w));
44     for k = 1:length(w)
45         Xw(k) = sum(x .* exp(-1j * w(k) * n));
46     end
47 end
48

```



## EXERCISE#05

Documents ▸ MATLAB ▸ DSPLAB5

Editor - C:\Users\hasee\OneDrive\Documents\MATLAB\DSPLAB5\EXERCISE5.m

```
DTFT.m x EXERCISE2.m x EXERCISE3.m x EXERCISE4.m x EXERCISE5.m x EXERCISE7.m x POSTLAB.m x +
1      x = [1 1 1];
2      n = -1:1;
3
4      w = 0:pi/100:2*pi;
5
6      Xw = dtft(x, n, w);
7
8      E_time = sum(abs(x).^2);
9
10     E_freq = trapz(w, abs(Xw).^2) / (2 * pi);
11
12     fprintf('Energy computed from time domain: %.4f\n', E_time);
13     fprintf('Energy computed from frequency domain: %.4f\n', E_freq);
14
15
16     function Xw = dtft(x, n, w)
17         Xw = zeros(size(w));
18         for k = 1:length(w)
19             Xw(k) = sum(x .* exp(-1j * w(k) * n));
20         end
21     end
22
```

### Command Window

New to MATLAB? See resources for [Getting Started](#).

```
>> EXERCISE5
```

```
Energy computed from time domain: 3.0000
```

```
Energy computed from frequency domain: 3.0000
```

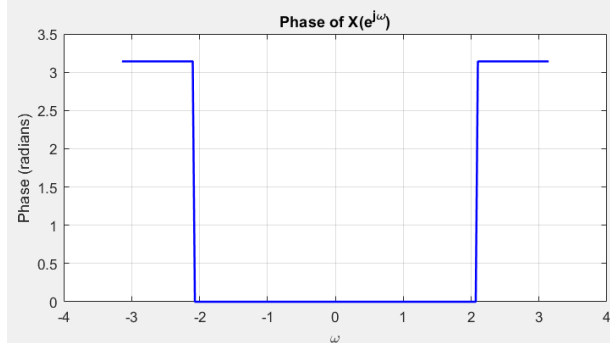
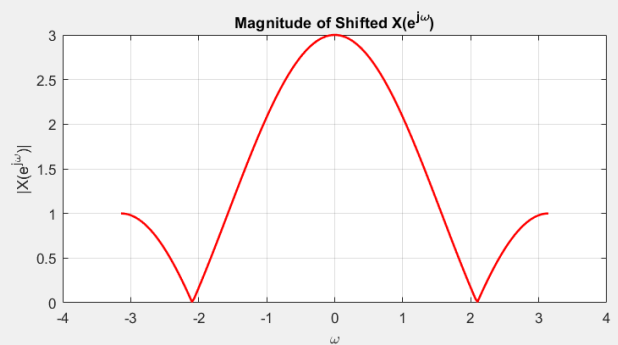
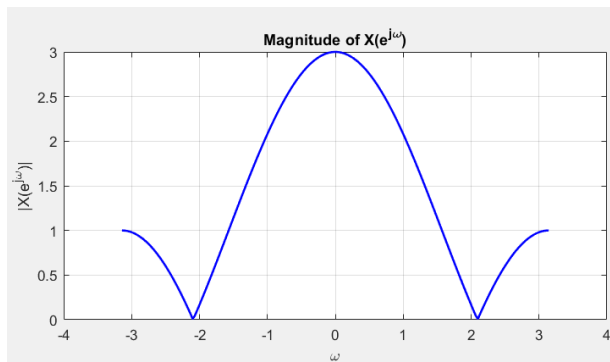
```
fx >>
```

## EXERCISE#06

```

Editor - C:\Users\hasee\OneDrive\Documents\MATLAB\DSI
DTFT.m  EXERCISE2.m  EXERCISE3.m  EX
1      x = [1 1 1];
2      n = -1:1;
3
4      w = linspace(-pi, pi, 200);
5
6      xw = dtft(x, n, w);
7      d = -4;
8      n_shifted = n + d;
9      xw_shifted = xw .* exp(-1j * w * d);
10
11     figure;
12     subplot(2,2,1);
13     plot(w, abs(xw), 'b', 'Linewidth', 1.5);
14     title('Magnitude of X(e^{j\omega})');
15     xlabel('\omega');
16     ylabel('|X(e^{j\omega})|');
17     grid on;
18
19     subplot(2,2,2);
20     plot(w, abs(xw_shifted), 'r', 'Linewidth', 1.5);
21     title('Magnitude of Shifted X(e^{j\omega})');
22     xlabel('\omega');
23     ylabel('|X(e^{j\omega})|');
24     grid on;
25
26     subplot(2,2,3);
27     plot(w, angle(xw), 'b', 'Linewidth', 1.5);
28     title('Phase of X(e^{j\omega})');
29     xlabel('\omega');
30     ylabel('Phase (radians)');
31     grid on;
32
33     subplot(2,2,4);
34     plot(w, angle(xw_shifted), 'r', 'Linewidth', 1.5);
35     title('Phase of Shifted X(e^{j\omega})');
36     xlabel('\omega');
37     ylabel('Phase (radians)');
38     grid on;
39     function xw = dtft(x, n, w)
40         xw = zeros(size(w));
41         for k = 1:length(w)
42             xw(k) = sum(x .* exp(-1j * w(k) * n));
43         end
44     end
45

```



## POST LAB:

Documents > MATLAB > DSPLAB3

```

Editor - C:\Users\hasee\OneDrive\Documents\MATLAB\DSPLAB3
DTFT.m x EXERCISE2.m x EXERCISE3.m x EXERCISE4.m x

1  x = [1/3, 1/3, 1/3];
2  n = [0 1 2];
3  w = linspace(-pi, pi, 200);
4  xw = dtft(x, n, w);
5
6  x_conj = conj(x);
7  xw_conj = dtft(x_conj, n, w);
8
9  xw_neg = conj(Flip(xw));
10
11 figure;
12 subplot(2,2,1);
13 plot(w, abs(xw), 'b', 'Linewidth', 1.5);
14 title(' |X(e^{j\omega})| ');
15 xlabel('\omega');
16 ylabel('Magnitude');
17 grid on;
18
19 subplot(2,2,2);
20 plot(w, abs(xw_conj), 'r', 'Linewidth', 1.5);
21 title(' |X^*(e^{j\omega})| ');
22 xlabel('\omega');
23 ylabel('Magnitude');
24 grid on;
25
26 subplot(2,2,3);
27 plot(w, angle(xw), 'b', 'Linewidth', 1.5);
28 title('Phase of X(e^{j\omega})');
29 xlabel('\omega');
30 ylabel('Phase (radians)');
31 grid on;
32
33 subplot(2,2,4);
34 plot(w, angle(xw_neg), 'r', 'Linewidth', 1.5);
35 title('Phase of X^*(-\omega)');
36 xlabel('\omega');
37 ylabel('Phase (radians)');
38 grid on;
39
40 function xw = dtft(x, n, w)
41     xw = zeros(size(w));
42     for k = 1:length(w)
43         xw(k) = sum(x .* exp(-1j * w(k) * n));
44     end
45 end
46

```

